

1	YTSS/2025/I/24
	A tuition centre offers basic, intermediate and advanced English writing courses. Each term is made up of a 11-week block of sessions with one session per week. The matrix L shows the number of students enrolled in each session. $L = \begin{matrix} & \begin{matrix} B & I & A \end{matrix} \\ \begin{pmatrix} 10 & 15 & 8 \\ 16 & 20 & 12 \end{pmatrix} & \begin{matrix} \text{Weekday} \\ \text{Weekend} \end{matrix} \end{matrix}$ (a) Evaluate the matrix $M = 11L$. [1] (b) The centre charges \$40 for each basic session, \$45 for each intermediate session and \$50 for each advanced session. Represent these amounts in a 3×1 column matrix N. □ [1] (c) Evaluate the matrix $T = MN$. [2] (d) State what each of the elements of T represents. [1] (e) Due to a circuit breaker measure during a pandemic, the centre has to offer online class sessions instead. For the next 11-week block, the prices of the class sessions are reduced by 20%. The same number of students signed up. Using matrix multiplication, calculate the total amount of money the centre receives for the 11-week block of online English writing sessions. [2]

2	SCGS/2025/I/22						
	A small business makes jewellery. Workers are paid a basic rate with additional payment for each item they make as shown below. <table border="1"> <tr> <td>Basic rate</td><td>\$9.80 per hour</td></tr> <tr> <td>Necklace</td><td>\$1.65 each</td></tr> <tr> <td>Bracelet</td><td>\$1.45 each</td></tr> </table> (a) Represent the rates in a 3×1 column matrix R. [1] (b) Evaluate the matrix D = WR. [1] (c) Explain what the element in the second row of matrix D represents. [1] (d) Calculate how many more bracelets Betty would have to make to earn at least the same amount as Annie. [2]	Basic rate	\$9.80 per hour	Necklace	\$1.65 each	Bracelet	\$1.45 each
Basic rate	\$9.80 per hour						
Necklace	\$1.65 each						
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3	CCHY/2025/II/05						
	Salim owns three cafes A, B and C which sell burgers and salad. The cafes operate 7 days a week. The sales on a weekend are on average half of that on a weekday. (a) Suggest a reason why sales on a weekend are lower than sales on a weekday. [1] (b) Evaluate the matrix W = $6\mathbf{F}$. [1] (c) In all of his three cafes, each burger costs \$12 and each plate of salad costs \$8. Represent the costs of the food items in a 2×1 column matrix C. [1] (d) Evaluate the matrix R = WC. [2] (e) State what each of the elements of R represents. [1] (f) All the food items are sold for 40% profit. By writing down a row matrix T, find TR which gives the total amount of profit all three cafes made in a week. [2]						
4	SST/2025/I/09						
	Diana has two outlets that sell donuts of three flavours. The number of donuts sold in the outlets over a two-hour period is given by the matrix D. $\mathbf{D} = \begin{pmatrix} \text{Blueberry} & \text{Chocolate} & \text{Apple} \\ 75 & 86 & 48 \\ 108 & 56 & 36 \end{pmatrix} \begin{matrix} \text{Outlet 1} \\ \text{Outlet 2} \end{matrix}$ (a) Each blueberry donut is sold at \$3.50. Each chocolate donut is sold at \$4.50. Each apple donut is sold at \$2.50. Represent the selling price in a 3×1 column matrix P. [1] (b) Evaluate the matrix M = DP. [1] (c) Explain what each element in matrix M represents. [1]						

5 GMSS/2025/II/03

The table below shows the price of a ticket for each category for a musical, Swan Lake.

Child (below 12 years old)	\$8
Adult	\$30
Senior Citizen (above 55 years old)	\$20

(a) Represent the above information in a 3×1 matrix **A**. [1]

(b) Mrs Lim, aged 40, bought four tickets for her 70 year-old father, her 9 year-old and 11 year-old daughters, and herself.

(i) Write down a matrix **B** such that the product **BA** gives the total amount of money Mrs Lim paid for the tickets. [1]

(ii) Evaluate the matrix **C** = **BA**. [2]

(c) The table below shows the number of tickets sold for the musical on Saturday and Sunday.

Number of tickets sold

	Child	Adult	Senior Citizen
Saturday	37	u	25
Sunday	44	85	v

The total revenue from the sale of tickets collected on Saturday and Sunday are \$2686 and \$3522 respectively.

Using matrix multiplication, find the value of u and of v . [3]

(d) Evaluate the matrix **P** = $\begin{pmatrix} 1.2 & 2 \end{pmatrix}$ **T** where **T** is the matrix representing the revenue collected on Saturday and Sunday. [1]

(e) Explain what the elements in $\begin{pmatrix} 1.2 & 2 \end{pmatrix}$ represent. [1]

6 ACSBR/2025/I/21

In hardware store *A*, a plier cost \$8.90, a spanner cost \$6.50 and a hammer cost \$9.90.

The same items can be bought in hardware store *B*.

In hardware store *B*, the plier cost \$7.90, the spanner cost \$7.10 and the hammer cost \$ x .

This information can be represented by the matrix **P** = $\begin{pmatrix} 8.90 & 7.90 \\ 6.50 & 7.10 \\ 9.90 & x \end{pmatrix}$

(a) Lee buys 3 pliers, 5 spanners and 2 hammers.
Lin buys y pliers, 3 spanners and 4 hammers.

Represent this information in a 2×3 matrix **Q**. [1]

(b) Find, in terms of x and y , the matrix **T** = **QP**. [2]

	(c) State what the elements in matrix T represents. [1] (d) Lee would save \$2.20 by buying his purchases from hardware store <i>A</i>. (i) Find the value of <i>x</i>. [1] (ii) Lin claims that it will cost him the same to buy all the things from hardware store <i>A</i> or <i>B</i>. Explain whether Lin is right. Show your workings clearly. [2]												
7	KRSS/2025/I/24 The table below shows the number of \$5 and \$10 notes that Ali, Rin and Ryu has. <table><tr><td></td><td>\$5</td><td>\$10</td></tr><tr><td>Ali</td><td>4</td><td>3</td></tr><tr><td>Rin</td><td>0</td><td>6</td></tr><tr><td>Ryu</td><td>2</td><td>5</td></tr></table> The information in the table can be represented by a 3 by 2 matrix P. (a) Write down the matrix P. [1] (b) Calculate PS where $S = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$. [1] (c) Describe what is represented by the elements of PS. <i>Answer</i> [1] (d) By using matrix multiplication, calculate the total money that Ali, Rin and Ryu has. [2]		\$5	\$10	Ali	4	3	Rin	0	6	Ryu	2	5
	\$5	\$10											
Ali	4	3											
Rin	0	6											
Ryu	2	5											

Solution

1	(a) $M = \begin{pmatrix} 110 & 165 & 88 \\ 176 & 220 & 132 \end{pmatrix}$ (b) $N = \begin{pmatrix} 40 \\ 45 \\ 50 \end{pmatrix}$ (c) $= \begin{pmatrix} 16225 \\ 23540 \end{pmatrix}$ (d) It represents that the total amount the centre collected for <u>basic, intermediate and advanced English class in a term (or 11 weeks) for weekday and weekend classes respectively.</u> --- [B1] (e) $(0.8 \quad 0.8) \begin{pmatrix} 16225 \\ 23540 \end{pmatrix} \text{---[M1]}$ $= (31812)$ \$31812 ---[A1]
2	(a) $R = \begin{pmatrix} 9.80 \\ 1.65 \\ 1.45 \end{pmatrix}$ (b) $D = \begin{pmatrix} 142.6 \\ 139.05 \end{pmatrix}$ (c) It represents the total amount Betty was paid or earned. (d) Difference = $142.6 - 139.05 = 3.55$ $1.45x \geq 3.55$ $x \geq \frac{3.55}{1.45} \Rightarrow x \geq 2\frac{13}{29}$ Betty must make 3 more bracelets.

3

(a)

The cafés could be located in the CBD/business area/school area with more workers/students visiting the cafés for meals on the weekdays.

OR

the main clientele of the cafes are company workers and students who would get their burgers and salads during the weekdays.

(b)

$$W = \begin{pmatrix} 720 & 66 \\ 72 & 810 \\ 480 & 552 \end{pmatrix}$$

(c)

$$C = \begin{pmatrix} 12 \\ 8 \end{pmatrix}$$

(d)

$$R = \begin{pmatrix} 9168 \\ 7344 \\ 10176 \end{pmatrix}$$

(e)

The elements represent the amount of money collected from the sales of burgers and salads from café A, B and C respectively in a week.

(f)

$$\begin{aligned} TR &= \begin{pmatrix} 0.4 & 0.4 & 0.4 \\ 1.4 & 1.4 & 1.4 \end{pmatrix} \begin{pmatrix} 9168 \\ 7344 \\ 10176 \end{pmatrix} \\ &= \frac{2}{7}(9168 + 7344 + 10176) \\ &= (7625.14) \text{ (2 d.p.)} \end{aligned}$$

4

(a)

$$P = \begin{pmatrix} 3.5 \\ 4.5 \\ 2.5 \end{pmatrix}$$

(b)

$$M = DP$$

$$\begin{aligned} &= \begin{pmatrix} 75 & 86 & 48 \\ 108 & 56 & 36 \end{pmatrix} \begin{pmatrix} 3.5 \\ 4.5 \\ 2.5 \end{pmatrix} \\ &= \begin{pmatrix} (75 \times 3.5) + (86 \times 4.5) + (48 \times 2.5) \\ (108 \times 3.5) + (56 \times 4.5) + (36 \times 2.5) \end{pmatrix} \\ &= \begin{pmatrix} 769.5 \\ 720 \end{pmatrix} \end{aligned}$$

(c)

769.5 represents the total amount in dollars collected by Diana from selling donuts of the 3 flavours in outlet 1 over a two-hour period.

720 represents the total amount collected in dollars by Diana from selling donuts of the 3 flavours in outlet 2 over a two-hour period.

5	3a	$A = \begin{pmatrix} 8 \\ 30 \\ 20 \end{pmatrix}$
	3bi	$B = \begin{pmatrix} 2 & 1 & 1 \end{pmatrix}$
	3bii	$C = \begin{pmatrix} 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 30 \\ 20 \end{pmatrix}$ $= (66)$
	3c	$\begin{pmatrix} 37 & u & 25 \\ 44 & 85 & v \end{pmatrix} \begin{pmatrix} 8 \\ 30 \\ 20 \end{pmatrix} = \begin{pmatrix} 2686 \\ 3522 \end{pmatrix}$ $\begin{pmatrix} 296 + 30u + 500 \\ 352 + 2550 + 20v \end{pmatrix} = \begin{pmatrix} 2686 \\ 3522 \end{pmatrix}$ $30u = 1890 \quad \text{and} \quad 20v = 620$ $u = 63 \quad \quad \quad v = 31$
	3d	$P = \begin{pmatrix} 1.2 & 2 \end{pmatrix} \begin{pmatrix} 2686 \\ 3522 \end{pmatrix}$ $= (10267.2)$
	3e	1.2 means total revenue increase by 20% on Saturday and 2 means total revenue increase by 100% on Sunday.
6	(a)	$Q = \begin{pmatrix} 3 & 5 & 2 \\ y & 3 & 4 \end{pmatrix}$
	(b)	$T = \begin{pmatrix} 3 & 5 & 2 \\ y & 3 & 4 \end{pmatrix} \begin{pmatrix} 8.90 & 7.90 \\ 6.50 & 7.10 \\ 9.90 & x \end{pmatrix} = \begin{pmatrix} 79 & 59.2 + 2x \\ 8.9y + 59.1 & 7.9y + 21.3 + 4x \end{pmatrix}$
	(c)	Each element represents the total cost of Lee's and Lin's purchase from each of the shops A & B respectively.
	di)	$79 + 2.2 = 59.2 + 2x$ $x = 11$
	dii)	$8.9y + 59.1 = 7.9y + 21.3 + 4(11)$ $y = 6.2$
		It is not possible to buy a non-whole number plier. Therefore, Lin's claim will not be right.

7	a	$\begin{pmatrix} 4 & 3 \\ 0 & 6 \\ 2 & 5 \end{pmatrix}$
	b	$\begin{pmatrix} 50 \\ 60 \\ 60 \end{pmatrix}$
	c	The total money Ali, Rin and Ryu has respectively
	d	$(1 \ 1 \ 1) \begin{pmatrix} 50 \\ 60 \\ 60 \end{pmatrix}$ (170) \$170